class18

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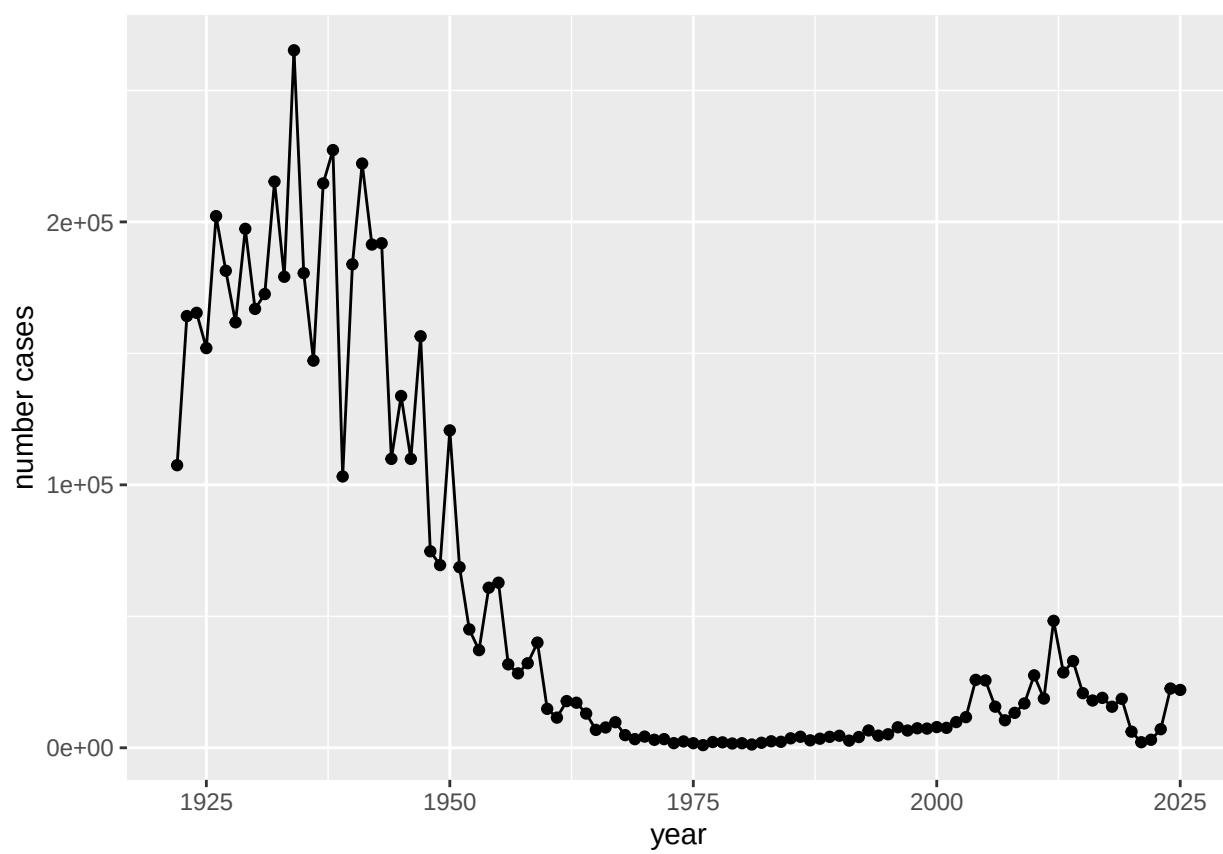
##bg pertussis aka whooping cough bac b.Pertussis can infect all ages, most severe for those under 1yr cdc tracking can be scraped w **datapasta** pkg

```
cdc<- data.frame(
                                          Year = c(1922L,1923L,1924L,
                                                   1925L,1926L,1927L,1928L,
                                                   1929L,1930L,1931L,1932L,
                                                   1933L,1934L,1935L,1936L,
                                                   1937L,1938L,1939L,1940L,
                                                   1941L,1942L,1943L,1944L,
                                                   1945L,1946L,1947L,1948L,
                                                   1949L,1950L,1951L,
                                                   1952L,1953L,1954L,1955L,
                                                   1956L,1957L,1958L,1959L,
                                                   1960L,1961L,1962L,1963L,
                                                   1964L,1965L,1966L,1967L,
                                                   1968L,1969L,1970L,1971L,
                                                   1972L,1973L,1974L,1975L,
                                                   1976L,1977L,1978L,
                                                   1979L,1980L,1981L,1982L,
                                                   1983L,1984L,1985L,1986L,
                                                   1987L,1988L,1989L,1990L,
                                                   1991L,1992L,1993L,1994L,
                                                   1995L,1996L,1997L,1998L,
                                                   1999L,2000L,2001L,2002L,
                                                   2003L,2004L,2005L,
                                                   2006L,2007L,2008L,2009L,
                                                   2010L,2011L,2012L,2013L,
                                                   2014L,2015L,2016L,2017L,
                                                   2018L,2019L,2020L,2021L,
                                                   2022L,2023L, 2024L, 2025L),
                     No..Reported.Pertussis.Cases = c(107473,164191,165418,
                                                   152003,202210,181411,
                                                   161799,197371,166914,172559,
                                                   215343,179135,265269,
                                                   180518,147237,214652,
                                                   227319,103188,183866,222202,
                                                   191383,191890,109873,
                                                   133792,109860,156517,74715,
                                                   69479,120718,68687,
                                                   45030,37129,60886,62786,
                                                   31732,28295,32148,40005,
                                                   14809,11468,17749,17135,
                                                   13005,6799,7717,9718,
```

```
                                                    4810,3285,4249,3036,3287,
                                                    1759,2402,1738,1010,
                                                    2177,2063,1623,1730,1248,
                                                    1895,2463,2276,3589,
                                                    4195,2823,3450,4157,4570,
                                                    2719,4083,6586,4617,
                                                    5137,7796,6564,7405,7298,
                                                    7867,7580,9771,11647,
                                                    25827,25616,15632,10454,
                                                    13278,16858,27550,18719,
                                                    48277,28639,32971,20762,
                                                    17972,18975,15609,18617,
                                                    6124,2116,3044,7063, 22538, 21996)
                  )
```

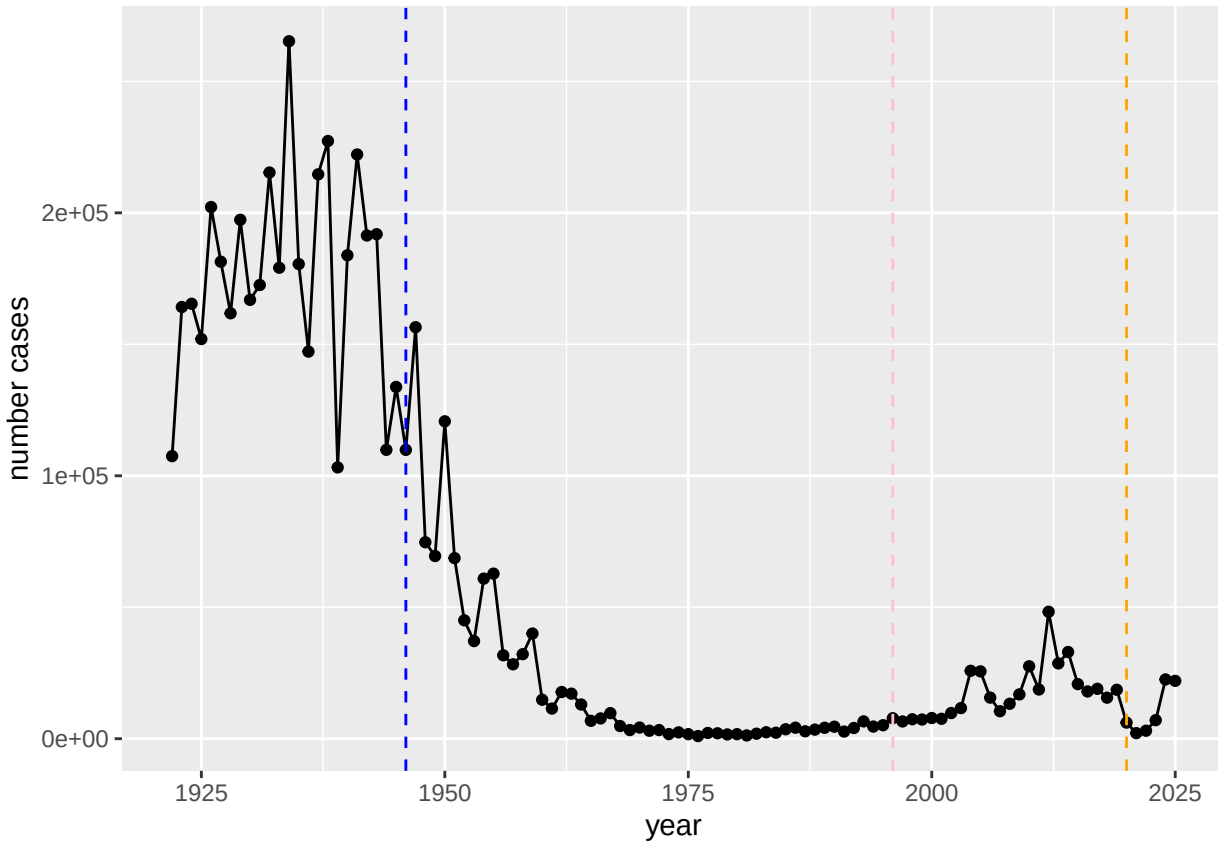
q1 year vs cases

```
library(ggplot2)
ggplot(cdc) +
  aes(Year, No..Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  labs(x="year", y="number cases")
```



>Q add major milestones inc first wp vax roll out (1946) switch to ap vax (1996) covid (2020)

```r
ggplot(cdc) +
  aes(Year, No..Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept= 1946, col= "blue", lty =2)+
  geom_vline(xintercept= 1996, col= "pink", lty =2)+
  geom_vline(xintercept= 2020, col= "orange", lty =2)+
  labs(x="year", y="number cases")
```



>high numbers pre 1940s, drop in 1950-1970s, but start rising before the 2000s, potentially due to the anti vax movement or just population just increasing or people not compliant with boosters? drop again after 2020, likely due to quarantine & no in-person school as a result of covid, then rise again before 2025, likely due to lockdown being lifted

> ** why upswing if vax** after ap vaccine drop, but get higher.... ap is safer but less effective so need boosters and people might not get the boosters, also better tests to detect

#cmipb proj comp models immunity json req api

```r
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject, 3)
```

```
##   subject_id infancy_vac biological_sex              ethnicity  race
## 1          1          wP         Female Not Hispanic or Latino White
```

```
## 2          2          wP         Female Not Hispanic or Latino White
## 3          3          wP         Female                Unknown White
##   year_of_birth date_of_boost      dataset
## 1    1986-01-01    2016-09-12 2020_dataset
## 2    1968-01-01    2019-01-28 2020_dataset
## 3    1983-01-01    2016-10-10 2020_dataset
```

Q how many ap wp in subject?

```r
table(subject$infancy_vac)
```

```
##
## aP wP
## 87 85
```

Q what is bio sex breakdown?

```r
table(subject$biological_sex)
```

```
##
## Female   Male
##    112     60
```

race bio sex? dataset representative of us pop?

```r
table(subject$race,subject$biological_sex)
```

```
##
##                                          Female Male
##   American Indian/Alaska Native               0    1
##   Asian                                      32   12
##   Black or African American                   2    3
##   More Than One Race                         15    4
##   Native Hawaiian or Other Pacific Islander   1    1
##   Unknown or Not Reported                    14    7
##   White                                      48   32
```

```r
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
head(specimen)
```

```
##   specimen_id subject_id actual_day_relative_to_boost
## 1           1          1                           -3
## 2           2          1                            1
## 3           3          1                            3
## 4           4          1                            7
## 5           5          1                           11
## 6           6          1                           32
##   planned_day_relative_to_boost specimen_type visit
## 1                             0         Blood     1
```

```
## 2                              1         Blood     2
## 3                              3         Blood     3
## 4                              7         Blood     4
## 5                             14         Blood     5
## 6                             30         Blood     6
```

use "join" middle tables so all data in one place, can use `*_join()` from **dplyr**

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
## Joining with `by = join_by(subject_id)`
```

```r
dim(meta)
```

```
## [1] 1503   13
```

```r
head(meta)
```

```
##   subject_id infancy_vac biological_sex              ethnicity  race
## 1          1          wP         Female Not Hispanic or Latino White
## 2          1          wP         Female Not Hispanic or Latino White
## 3          1          wP         Female Not Hispanic or Latino White
## 4          1          wP         Female Not Hispanic or Latino White
## 5          1          wP         Female Not Hispanic or Latino White
## 6          1          wP         Female Not Hispanic or Latino White
##   year_of_birth date_of_boost      dataset specimen_id
## 1    1986-01-01    2016-09-12 2020_dataset           1
## 2    1986-01-01    2016-09-12 2020_dataset           2
## 3    1986-01-01    2016-09-12 2020_dataset           3
## 4    1986-01-01    2016-09-12 2020_dataset           4
## 5    1986-01-01    2016-09-12 2020_dataset           5
## 6    1986-01-01    2016-09-12 2020_dataset           6
##   actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
## 1                           -3                             0         Blood
## 2                            1                             1         Blood
## 3                            3                             3         Blood
## 4                            7                             7         Blood
```

```
## 5                           11                            14         Blood
## 6                           32                            30         Blood
##   visit
## 1     1
## 2     2
## 3     3
## 4     4
## 5     5
## 6     6
```

```
abdata <- inner_join(titer, meta)
```

```
## Joining with `by = join_by(specimen_id)`
```

```
dim(abdata)
```

```
## [1] 61956    20
```

which isotypes?

```
table(abdata$isotype)
```

```
##
##   IgE   IgG  IgG1  IgG2  IgG3  IgG4
##  6698  7265 11993 12000 12000 12000
```

what reported?

```
table(abdata$antigen)
```

```
##
##     ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA
##    1970    1970    6318    1970    6712    6318    1970    1970    1970    6318
##     PD1     PRN      PT     PTM   Total      TT
##    1970    6712    6712    1970     788    6318
```

dataset? 2021 and 2022 are lower, 2023 is doubled.... 2021 and 2022 were affected by covid

```
table(abdata$dataset)
```

```
##
## 2020_dataset 2021_dataset 2022_dataset 2023_dataset
##        31520         8085         7301        15050
```

igg plot of mfi for all antigens

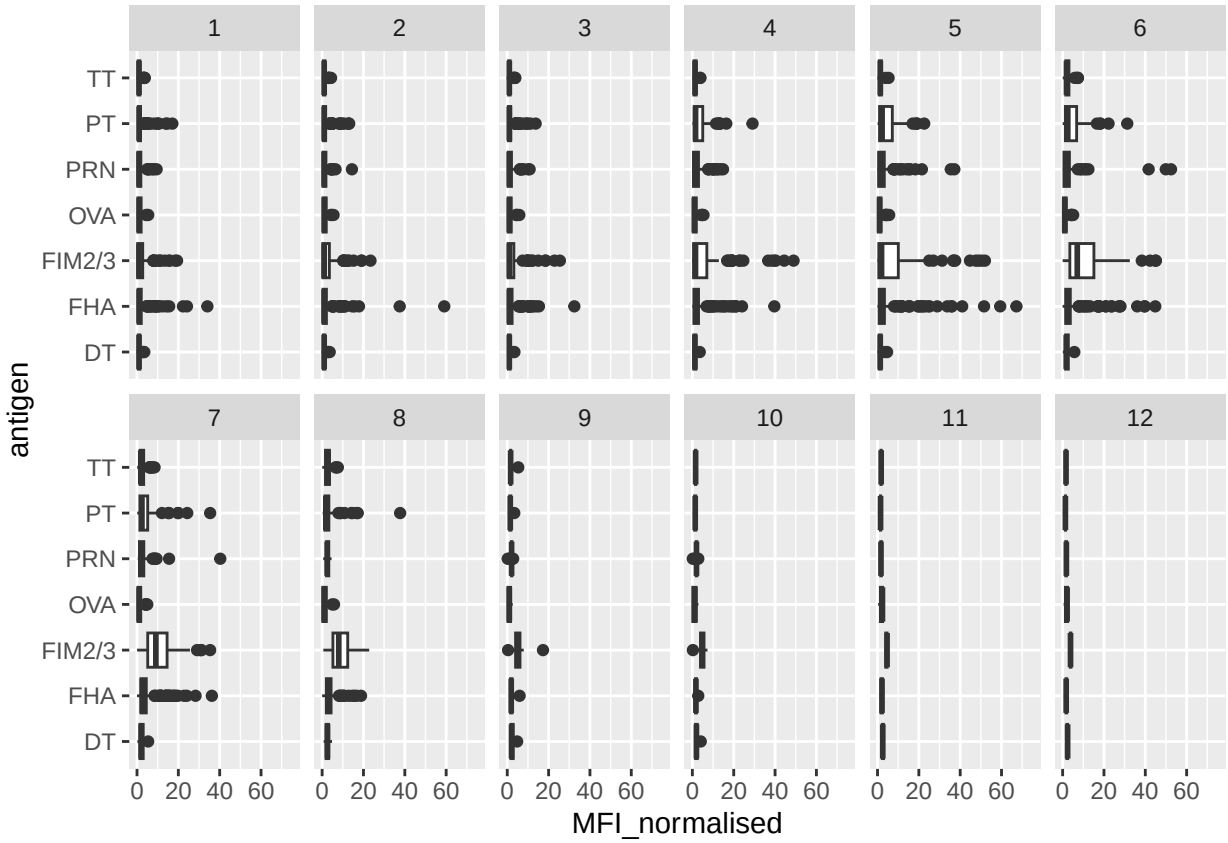
```
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

```
##   specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
## 1           1     IgG                TRUE      PT   68.56614       3.736992
## 2           1     IgG                TRUE     PRN  332.12718       2.602350
## 3           1     IgG                TRUE     FHA 1887.12263      34.050956
## 4          19     IgG                TRUE      PT   20.11607       1.096366
## 5          19     IgG                TRUE     PRN  976.67419       7.652635
## 6          19     IgG                TRUE     FHA   60.76626       1.096457
##    unit lower_limit_of_detection subject_id infancy_vac biological_sex
## 1 IU/ML                 0.530000          1          wP         Female
## 2 IU/ML                 6.205949          1          wP         Female
## 3 IU/ML                 4.679535          1          wP         Female
## 4 IU/ML                 0.530000          3          wP         Female
## 5 IU/ML                 6.205949          3          wP         Female
## 6 IU/ML                 4.679535          3          wP         Female
##                ethnicity  race year_of_birth date_of_boost      dataset
## 1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 4                Unknown White    1983-01-01    2016-10-10 2020_dataset
## 5                Unknown White    1983-01-01    2016-10-10 2020_dataset
## 6                Unknown White    1983-01-01    2016-10-10 2020_dataset
##   actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
## 1                           -3                             0         Blood
## 2                           -3                             0         Blood
## 3                           -3                             0         Blood
## 4                           -3                             0         Blood
## 5                           -3                             0         Blood
## 6                           -3                             0         Blood
##   visit
## 1     1
## 2     1
## 3     1
## 4     1
## 5     1
## 6     1
```

Q13 plot antigens

```r
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
    xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

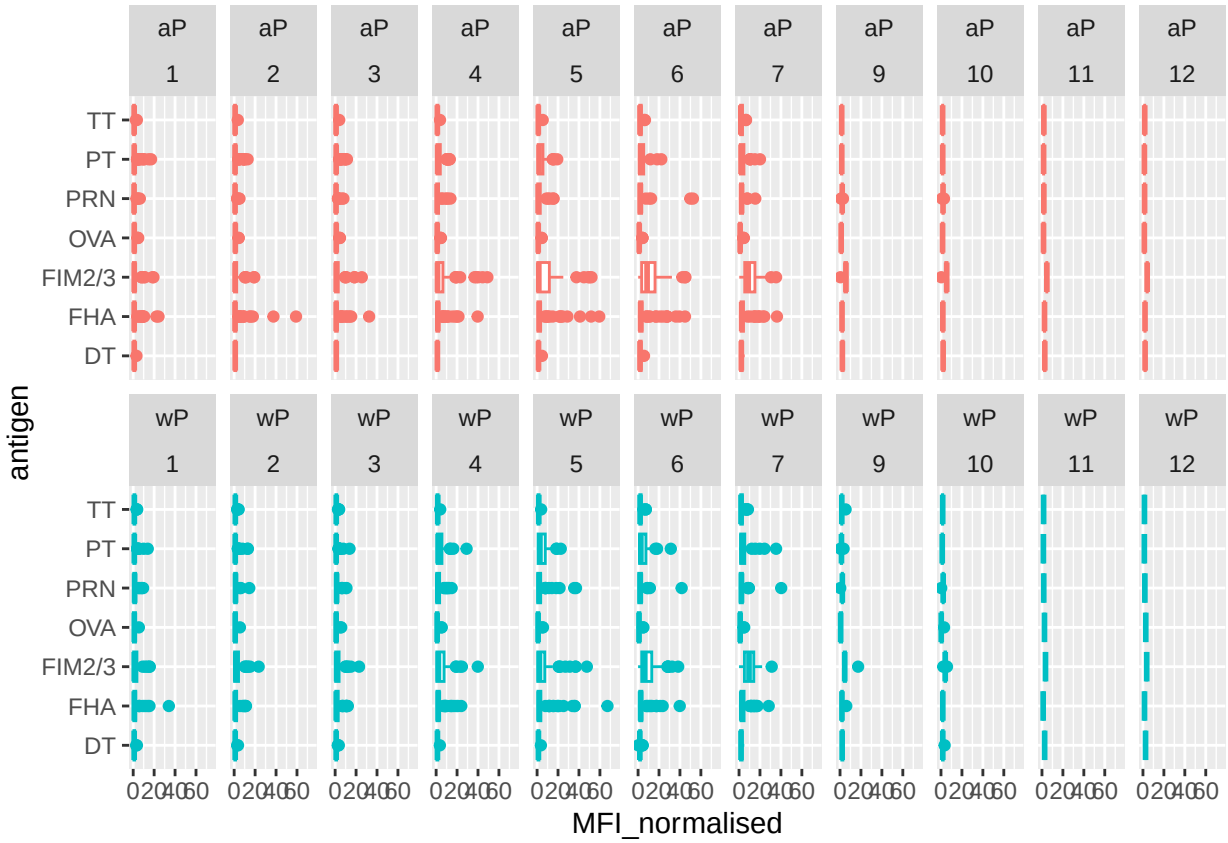
```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Q14 difference ap vs wp?

```
igg %>% filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)
```

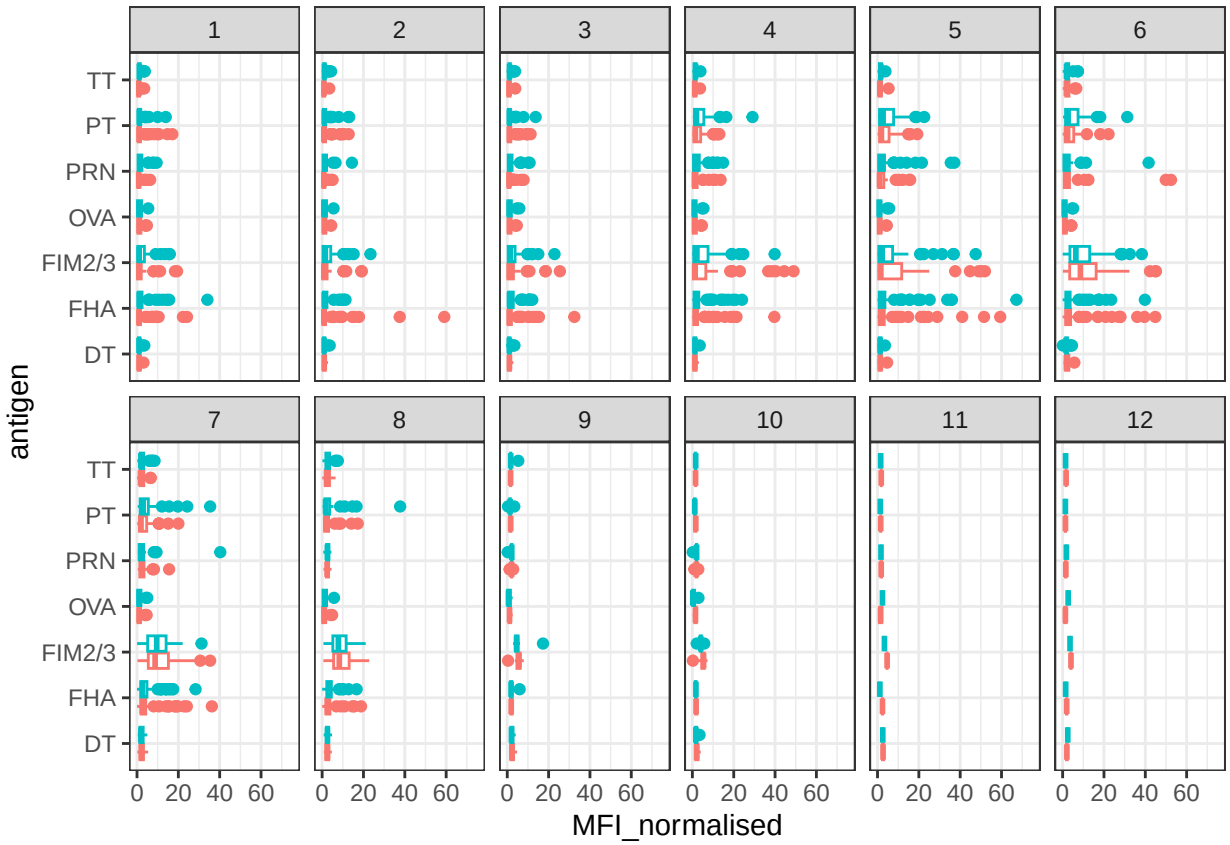
```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



>Qtemporal?

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

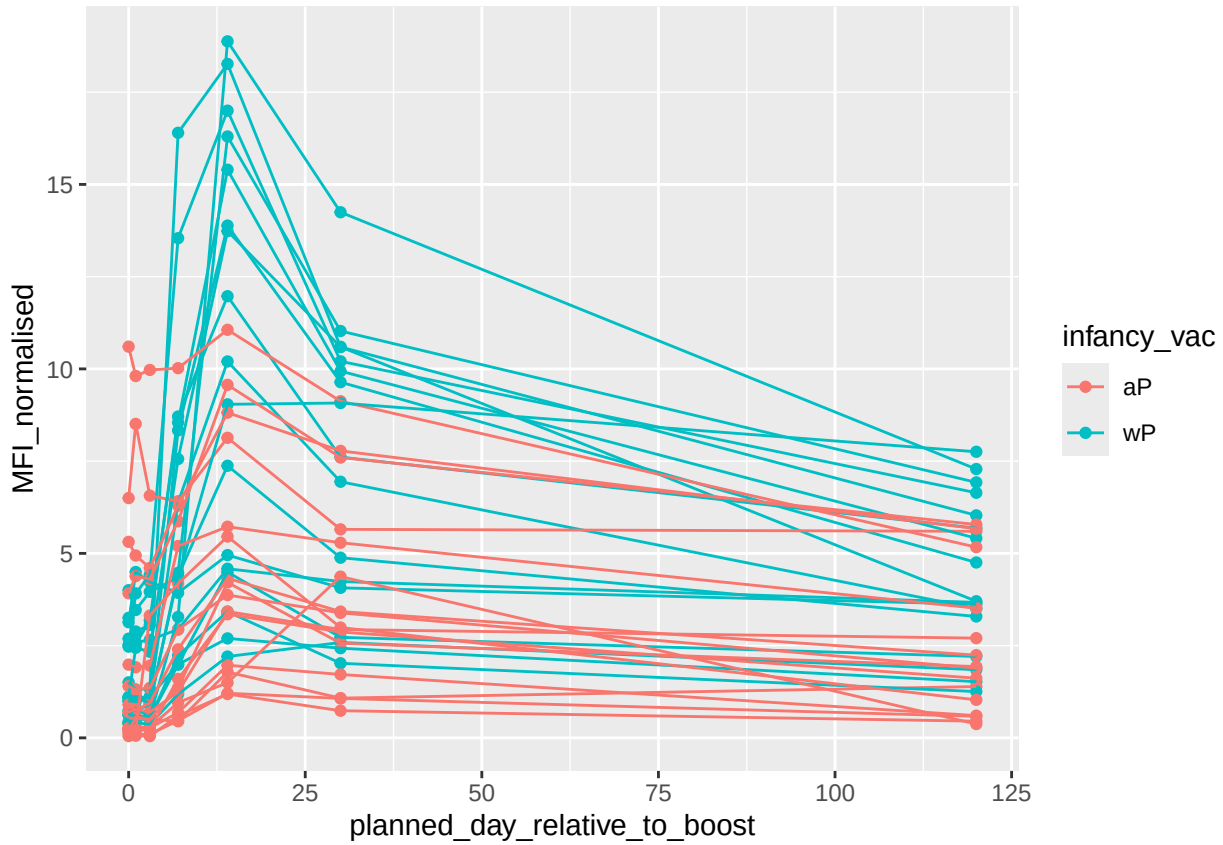
```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



focus on pt

```
pt.igg.21<- igg |> filter (antigen == "PT",
                           dataset == "2021_dataset")

ggplot(pt.igg.21)+
  aes(planned_day_relative_to_boost,
      MFI_normalised,
      col=infancy_vac,
      group=subject_id)+
  geom_point()+
  geom_line()
```

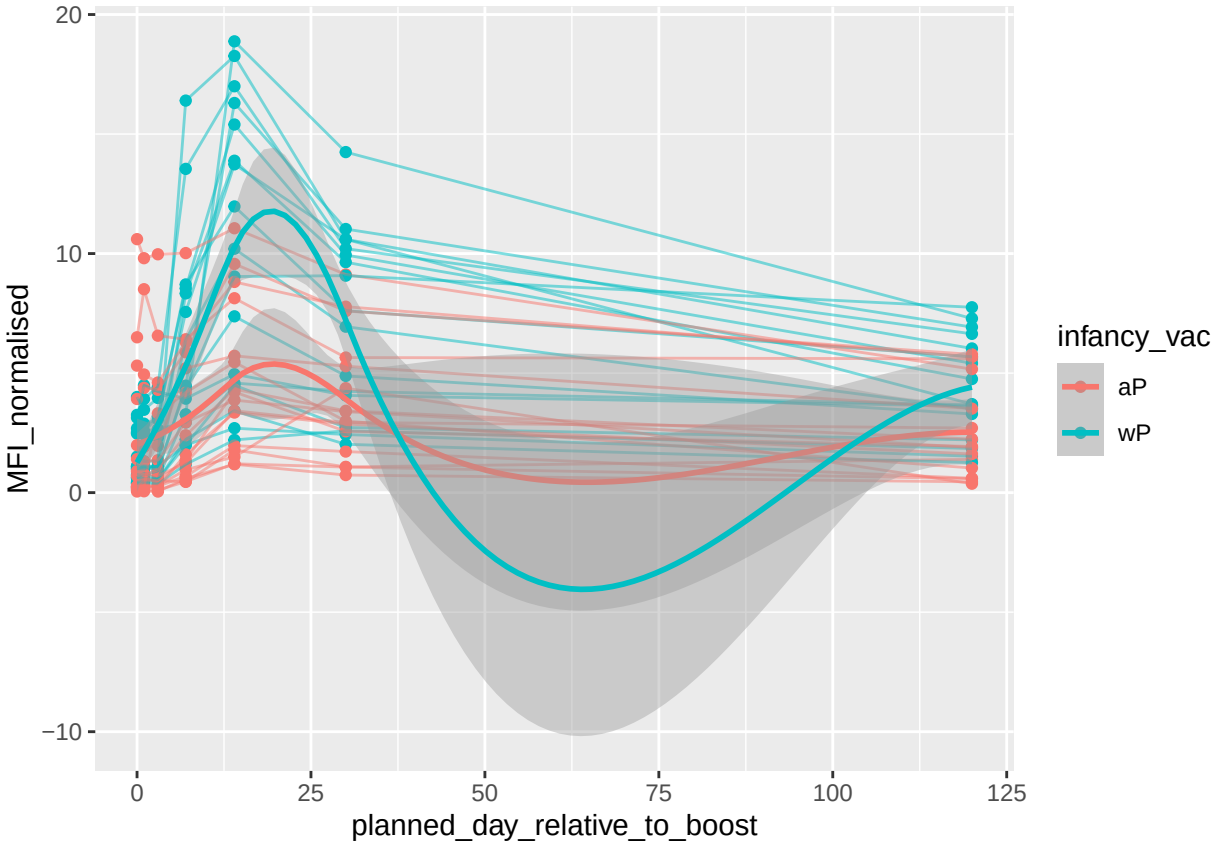


>Q geomsmooth

```
pt.igg.21<- igg |> filter (antigen == "PT",
                           dataset == "2021_dataset")

ggplot(pt.igg.21)+
  aes(planned_day_relative_to_boost,
      MFI_normalised,
      col=infancy_vac,
      group=subject_id)+
  geom_point()+
  geom_line(alpha=.5)+
  geom_smooth(aes(group = infancy_vac))
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```


20
10
0
−10
MFI_normalised
0
25
50
75
100
125
planned_day_relative_to_boost
infancy_vac
aP
wP